

Assignment - I

1) write about Network Hardware and Network Software

A: Network hardware

Network hardware refers to the physical devices used to connect computers and other devices so they can communicate and share data.

Ex:

- Router: Connects different networks and directs data b/w them.
- Switch: Connects devices within the same network and forwards data to the correct device.
- Hub: Connects multiple devices and sends data to all connected devices.
- Network Interface Card (NIC) - Enables a computer to connect to a network.
- Cables (Ethernet / Fiber Optic) - Carry data b/w network devices.

Network Software:

Network Software is the set of programs that manage, control, and secure communication b/w devices in a network.

Ex:

- Network operating system: manages network resources and users.
- Communication protocols: Such as TCP/IP, HTTP, FTP, and SMTP for data communication.

→ Network management Software :
monitors and manages network performance

→ Firewall Software :
protects the network performance

→ Antivirus Software : Detects and removes
malicious software.

② Discuss with neat sketch about OSI Reference model.

As OSI (Open Systems Interconnection) Reference model

→ The OSI Reference model is a 7-layer framework developed by the ISO (International Organization for Standardization) to standardize network communication. Each layer performs a specific function and communicates with the layers above and below it.

Neat Sketch

In OSI model

7. Application Layer

6. Presentation Layer

5. Session Layer

4. Transport Layer

3. Network Layer

2. Data link Layer

1. Physical Layer

→ Top-down approach
→ Bottom-up approach

Functions of Each layer

① Physical layer

→ Transmits raw bits over the communication medium.

→ Defines cables, connectors, signals, and transmission rate.

physical in three types

- ① establish ② maintain ③ Deactivate

functionalities - case

① Bit transmission / synchronization transmission

② Bit Rate Control

③ physical Topology

④ Transmission mode

- Simple mode
- Half duplex
- Full duplex

physical Layer

from datalink layer

L2 Data

10101100

physical Layer

To Datalink layer

L2 data

10101100

physical Layer

medium

② Data link Layer

- Transfers data b/w adjacent devices.
- performs error detection, flow control, and MAC addressing.
- Data represents in the form of frames

Sub layers of DLL

① Logical link Control layer

- It acts as an Interface to Network Layer
- Controls the flow

② Flow Control

③ Error Control

④ Framing

→ Headers

Sender & Receiver

DATA

Original Content

Trailers

- error control technique
- It checks Receiver address information

③ Network Layer

- Routes data from source to destination
- uses logical (IP) addressing and path selection.
- Host to Host delivery

functionalities

① Logical addressing

- Network layer uses IP address to transfer the data.

② Routing: Determines best path for data transfer using Routing algorithm & Routing Tables.

③ Internet looking:
Node — to — Node
Host — to — Host
Ex: LAN — LAN
LAN — WAN

④ Fragmentation & Reassembly

→ Dividing Large packets into Small packets.
→ Combines all packets together & Reads original data.

⑤ Congestion Control

→ Sometimes too much traffic in n/w, To reduce that Network Layer uses this to overcome & improves the performance.

④ Transport Layer

→ Ensures reliable end-to-end Communication
→ Performs Segmentation, error recovery, and flow Control.

→ protocols: TCP, UDP

Functionalities

① End to End delivery

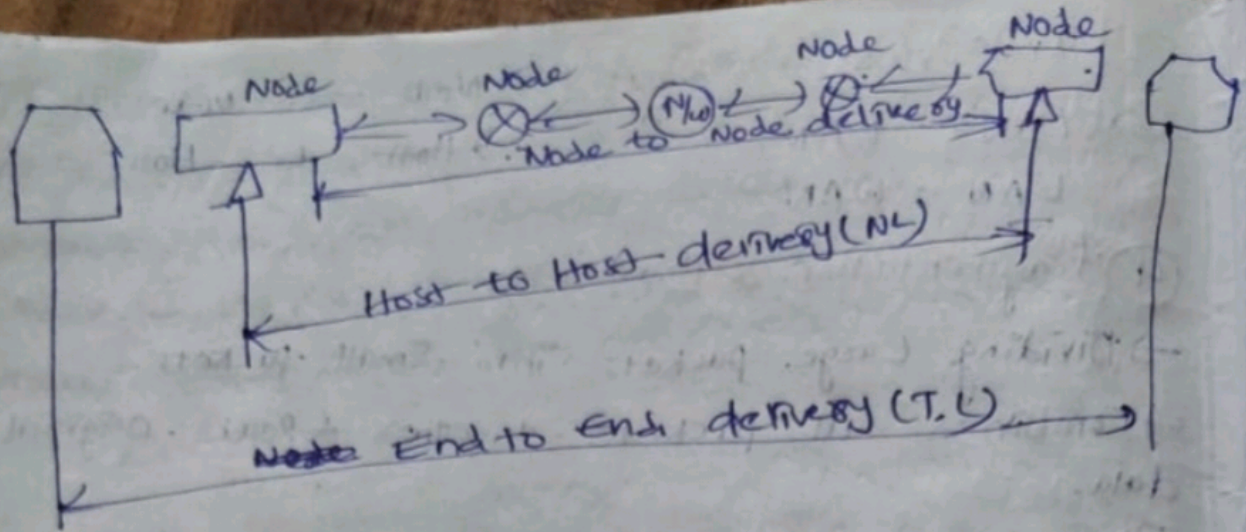
② Error Control

③ Flow Control

④ Access Control

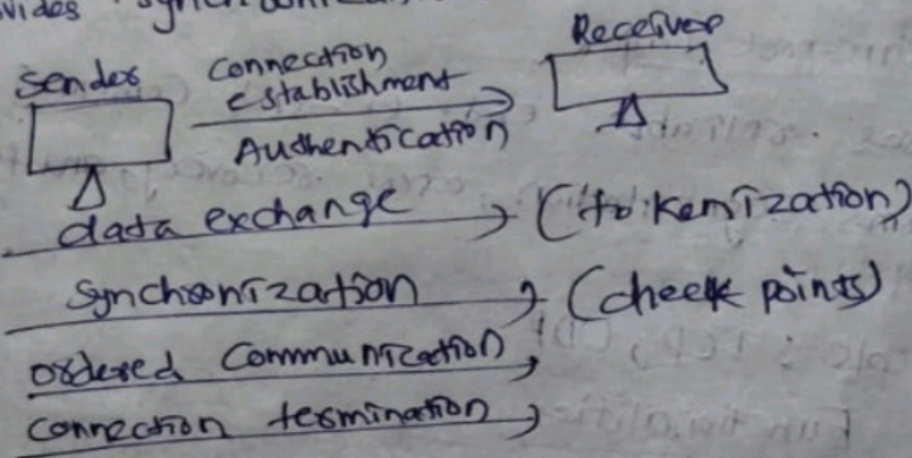
⑤ Connection oriented

⑥ Connectionless Services



⑤ Session Layer

- Establishes, manages, and terminates Communication Sessions.
- provides synchronization b/w devices.



Functionalities

- ① Connection establishment
- ② Connection maintenance
- ③ Synchronization
- ④ Dialog Control
- ⑤ Connection termination

⑥ Presentation Layer

- Translates data formats.
- performs data encryption, decryption, and data compression.
- So that it can be correctly understood by the application layer.

Functions:

- ① Data Translation: Converts data b/w different formats (e.g. - ASCII, Unicode)
- ② Encryption & Decryption: Secures data during transmission.
- ③ Data Compression: Reduces the size of data to improve transmission speed.

⑦ Application Layer

- provides network services directly to users.
- Ex. HTTP, FTP, SMTP, DNS

Functions

- provides an interface b/w users and the network.
- supports services such as email, file transfer, web browsing.
- provides network resource sharing.

③ Discuss with neat sketch about TCP/IP model

Ans The TCP/IP (Transmission Control Protocol / Internet Protocol) model is a 4-layer networking model used for communication over the Internet. It defines how data is transmitted from one device to another.

Neat Sketch

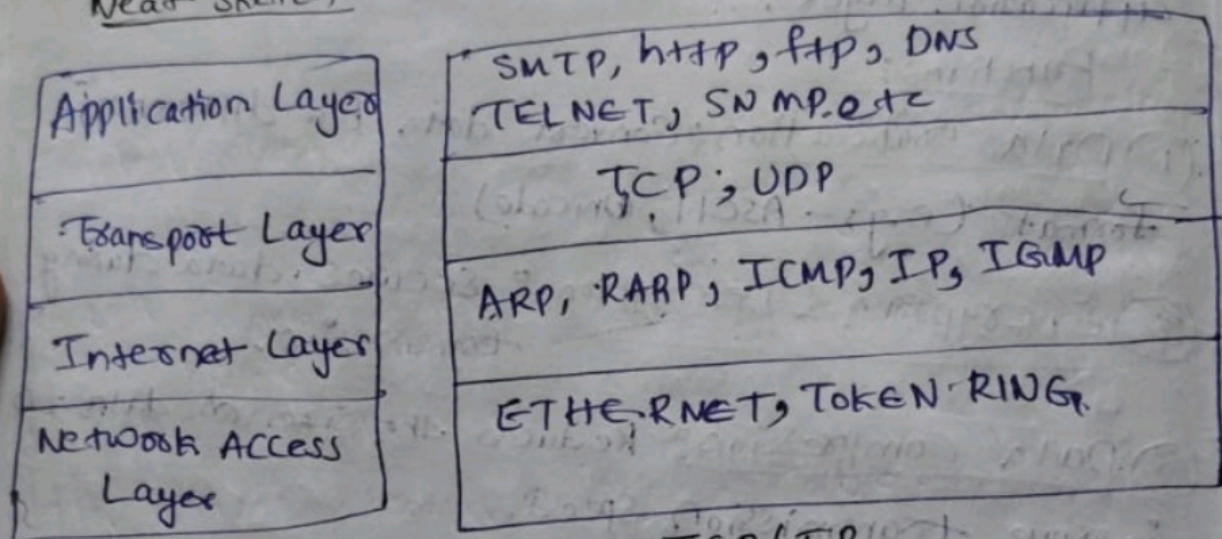


Fig 1 TCP/IP

TCP/IP
Fig; Protocol Stack

① Application Layer

- provides network services to users
- Combines the functions of the OSI Application, presentation, and session layers.

HTTP → used to load the website

SMTP → used for emails

FTP → download / upload files of a server

SNMP → monitoring & managing

② Transport Layer

- provides end-to-end communication.
- performs segmentation, error checking, and flow control
- protocols : TCP and UDP
- UDP → user datagram protocol { connectionless, fast
- TCP → Transmission Control protocol { connection-oriented, reliable

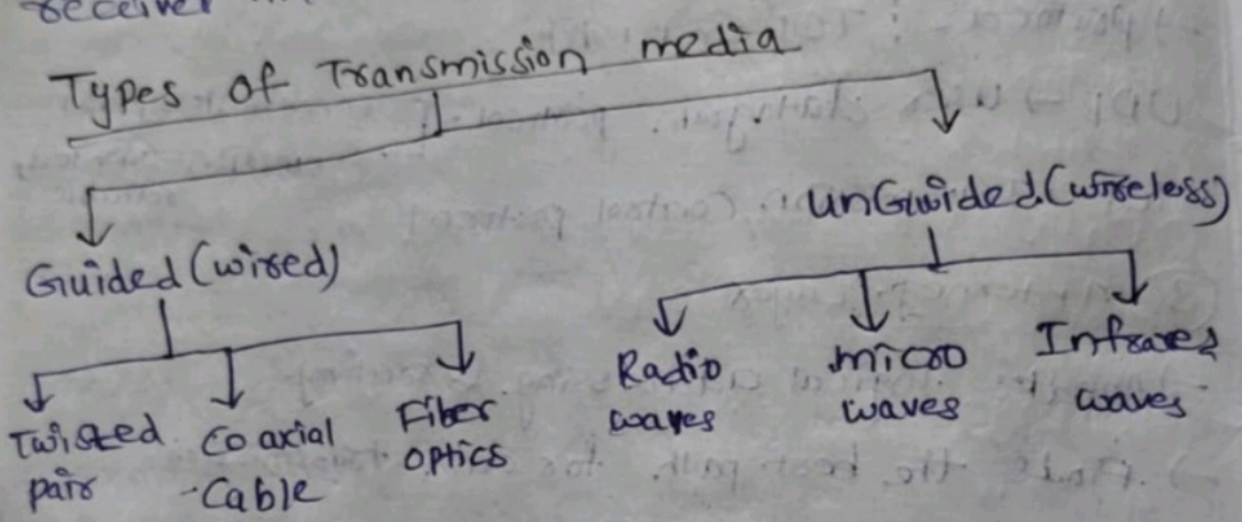
③ Internet Layer

- handles logical addressing & routing.
- Finds the best path for data transmission.
- protocols : IP, ICMP, ARP
- IP : internet protocol
- IPv4, IPv6
- ICMP → it used to check errors
- ARP { packet switching network
- TCP/IP

④ Network Access Layer

- Transfer data over the physical network.
- Handles framing, physical addressing, and transmission
- Ex: Ethernet, WiFi.

④ write about Transmission medium and types
As Transmission media is the communication path through which data travels from a sender to a receiver in a computer network.



① Guided (wired)

data travels through a physical cable.

Types:

- Twisted pair Cable & used in telephone lines and LANs
- Coaxial Cable & used in cable TV and broadband networks
- Optical Fiber Cable & uses light signals; provides very high speed & long-distance communication.

② Un Guided (wireless)

data travels through the air using electromagnetic waves.

Types:

- Radio waves - used in radio, broadcast and wifi
- Micro waves - used in satellite & mobile communication
- Infrared - used in TV remotes & short-range communication.

⑤ Explain about Data link layer — Design Issues

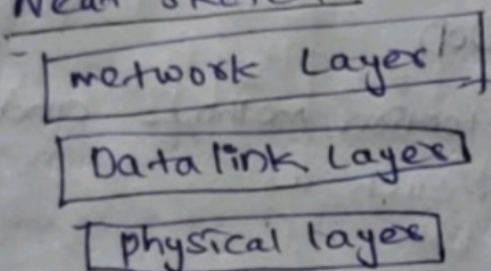
As ^{the} The Data link Layer is the 2nd layer of the OSI model.

→ Data represents in the form of frames

→ provides node-to-node delivery.

→ It provides reliable communication b/w two directly connected devices by organizing data into frames and detecting errors.

Neat sketch



Design Issues

① Framing

→ Divides data into frames for transmission

→ Adds header and trailer to each frame.

② physical Addressing

→ uses MAC (Media Access Control)

addresses to identify sender & receiver.

③ Error Control

→ Detects and corrects transmission errors.

→ uses error detection methods like CRC.

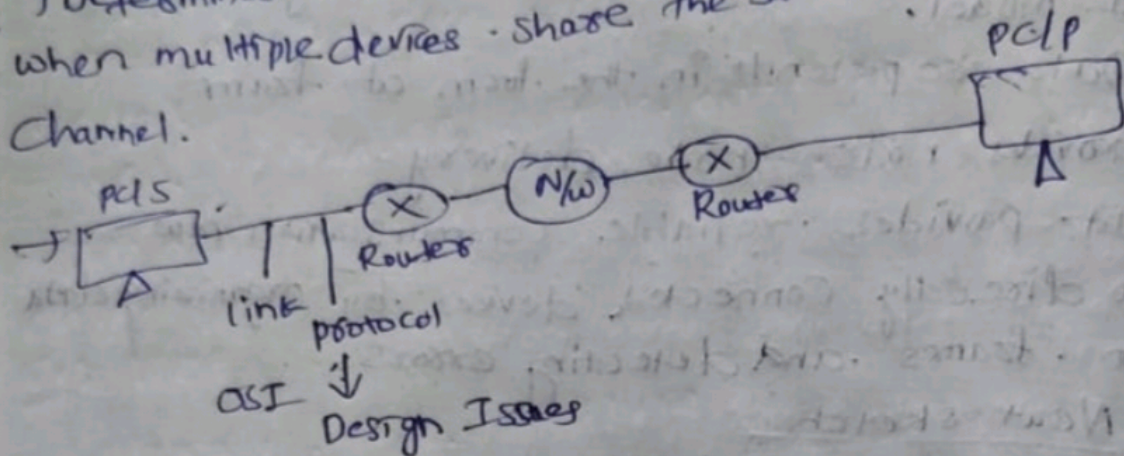
④ Flow Control

→ Controls the rate of data transmission so

the receiver is not overloaded.

⑤ Access Control

→ Determines which device can transmit data when multiple devices share the same communication channel.



TCP/IP developed by DoD

6) Explain about Error Detection methods and error correction methods. Such as

① parity check

② check sum

③ CRC

→ Error correction

Hamming Code

→ with an example

As Error detection : It is the process of identifying whether data has been corrupted during transmission

Error Correction : It is the process of detecting and correcting errors without retransmitting the data.

Error Detection methods

① Parity Check

- Adds one parity bit to the data.
- Can be Even parity or Odd parity.
- Detects single-bit errors.

Ex

- Data = 1011001 (4 ones → even)
- Even parity bit = 0
- Transmitted data = 10110010

② Check Sum

- Data is divided into blocks.
- The sender calculates the check sum & sends it with the data.
- The receiver recalculates the check sum. If both match, the data is accepted; otherwise, an error is detected.

③ Cyclic Redundancy Check (CRC)

- Uses polynomial division to generate a CRC value.
- The receiver performs the same calculation.
- If the remainder is zero, the data is considered error-free.
- CRC is widely used in Ethernet and communication networks.

Error Correction Method

Hamming Code

- one of the error detection and correction method
- used to identify single bit errors.
- It adds parity bits at specific positions in the data.

Ex: 1101

Hamming Code Structure

P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
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- Any length of data unit
power 2 = parity bit

$$2^x \geq m + x + 1$$

$$\text{if } x = 3$$

$$2^3 \geq 4 + 3 + 1$$

$$8 \geq 8$$

1101

$$4 + 3 + 7$$

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	1	0	0	1	1	0

$$P_1 = \text{check 1 skip 1} \quad P_2 = \text{check 2 skip 2}$$

$$= P_1 D_3 D_5 D_7 = P_1 \cdot P_2 D_5 \cdot D_6$$

$$= P_1 \cdot 1 \cdot 0 \cdot 1 = 0 \cdot P_2 \cdot 0 \cdot 1$$

$$P_1 = 0$$

$$P_2 = 1$$

$P_4 = \text{check } 4 \text{ skip } 4$

$$= P_1 \cdot P_2 \cdot D_3 \cdot P_4$$

$$= 0 \cdot 1 \cdot 1 \cdot P_4$$

$$= 0$$

$$\boxed{P_4 = 0}$$